

HW #1: Mixture Density Networks

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Probabilistic Deep Learning — Fall 2026

1 Model and implementation

For an input $\mathbf{x}$ and target $\mathbf{y}$, our mixture density network (MDN) models the conditional density as

$$p(\mathbf{y} | \mathbf{x}) = \sum_{k=1}^K \pi_k(\mathbf{x}) \mathcal{N}(\mathbf{y}; \boldsymbol{\mu}_k(\mathbf{x}), \boldsymbol{\Sigma}_k(\mathbf{x})), \quad \sum_{k=1}^K \pi_k(\mathbf{x}) = 1. \quad (1)$$

The network is trained by minimizing the mean negative log likelihood (NLL).

Describe the shared network, parameterization of mixture weights and scales, covariance assumptions, numerical-stability choices (for example, log-sum-exp and a minimum scale), optimizer, and hyperparameters. Explain how the deterministic baseline and the $K = 1$ Gaussian model differ.

2 Synthetic two-dimensional example

Define the synthetic input-output relationship, sample size, train/test split, noise process, and why $p(\mathbf{y} | \mathbf{x})$ is multimodal. State what behavior would validate the implementation.

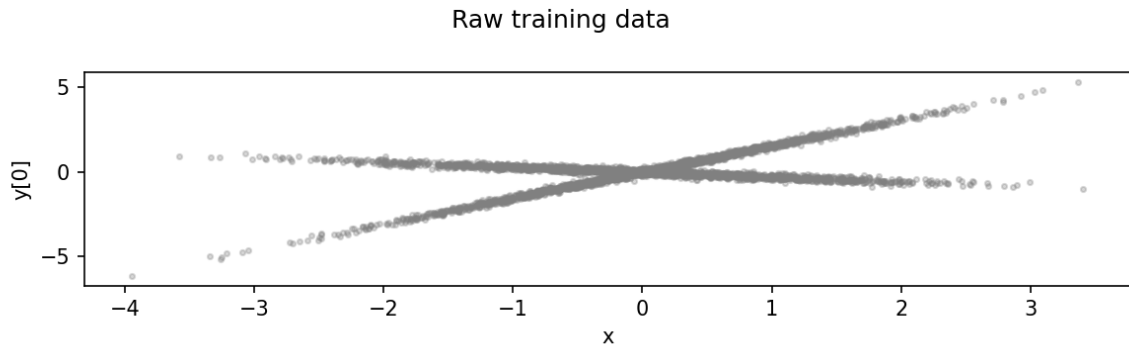


Figure 1: Raw training observations for the synthetic task, showing the multiple output modes associated with the input.

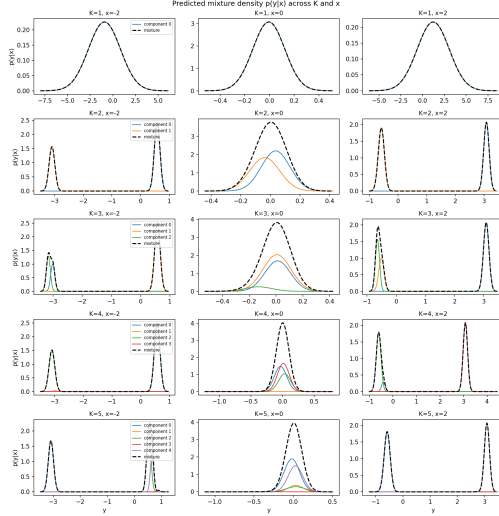


Figure 2: Learned conditional density for the synthetic 2D task over a grid of input and output values. The density should reveal the modes captured by the MDN and how their locations, weights, and widths vary with the input.

Table 1: Synthetic-data performance loaded directly from `csvs/synthetic_summary.csv`. NLL is not defined for the deterministic model.

Model	K	Train NLL	Test NLL	Train RMSE	Test RMSE
Deterministic	—	N/A	N/A	0.914497	0.920119
Single Gaussian	1	0.832337	0.807858	0.921334	0.905600
MDN	2	-0.259643	-0.265892	0.920106	0.907756
MDN	3	-0.263803	-0.265766	0.919739	0.908959
MDN	4	-0.262843	-0.265994	0.919391	0.909319
MDN	5	-0.262282	-0.258240	0.920974	0.905658

Interpret Figures 1 and 2. Discuss whether the MDN finds each mode, how mixture weights/scales change with the input, where the model fails, and whether the deterministic regressor averages incompatible outcomes.

3 Baselines and number of mixture components

We compare an ordinary deterministic regressor, a single-Gaussian network ($K = 1$), and MDNs with several values of K . All models use the same data split and, where possible, comparable capacity and training budgets.

Use Table 1 and Figure 3 to compare deterministic, $K = 1$, and $K > 1$ models. Explain the metrics, the benefit or cost of additional components, and any evidence of underfitting, overfitting, unstable components, or mode collapse.

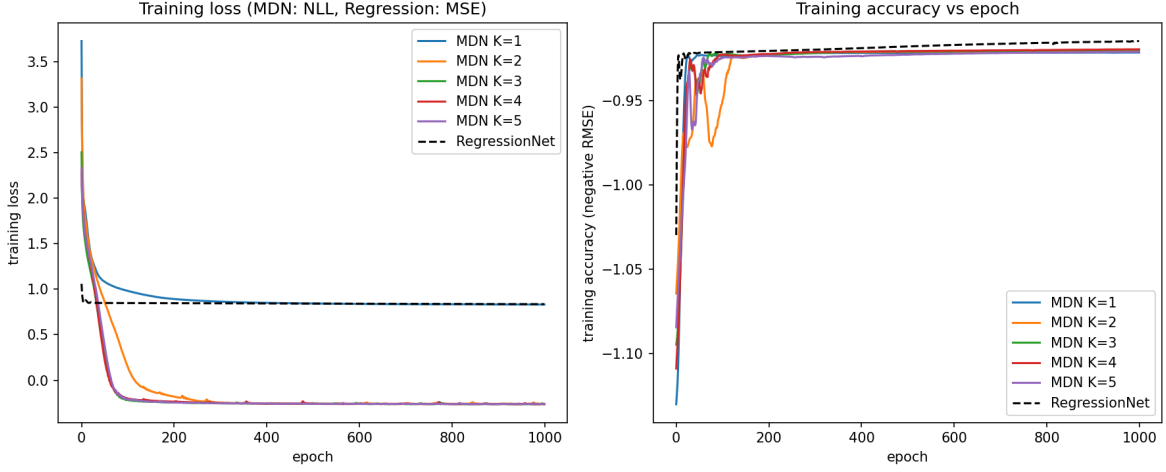


Figure 3: Training objective and RMSE across epochs for the deterministic baseline, single-Gaussian model, and MDNs with different numbers of components.

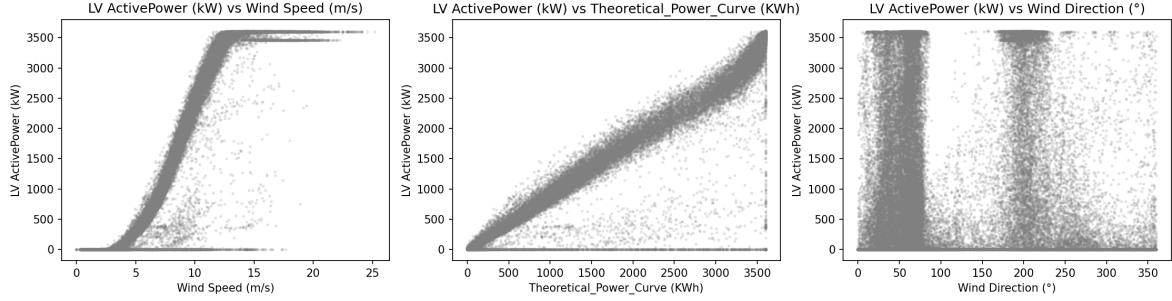


Figure 4: Wind-turbine active power against each available numeric predictor, used to assess multimodality and heteroscedasticity before fitting the models.

4 Higher-dimensional realistic data

We use the wind-turbine SCADA dataset as the realistic task. The response, input features, preprocessing, and split are specified below rather than assuming that adjacent time points are independent.

Describe the prediction target and all input dimensions. Summarize dataset size, missing/outlier handling, normalization, and the train/validation/ test protocol. Justify whether the split is random, grouped, or chronological, and explain why multimodal or heteroscedastic output may be plausible.

Compare training and test behavior on the realistic dataset. Discuss generalization gaps, predictive sharpness and calibration, regions of high uncertainty, the effect of K , and how conclusions differ from the controlled synthetic example. Reference Figures 4 and 5, Figure 6, and Table 2.

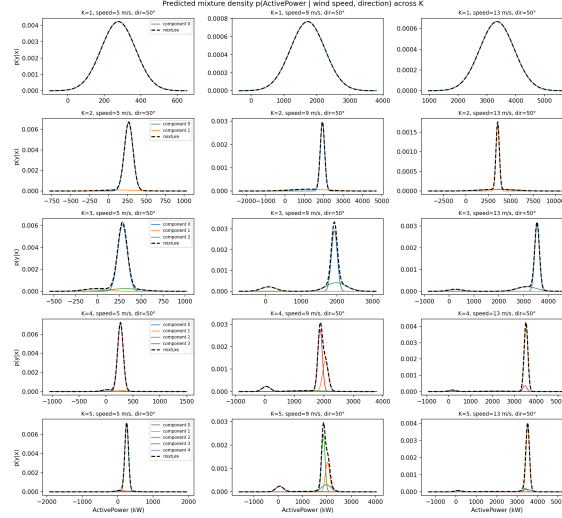


Figure 5: Learned conditional density for the wind-turbine task over a grid of inputs and active-power outputs. Use this view to show heteroscedasticity, multiple operating regimes, and regions where the model assigns diffuse or multimodal probability.

Table 2: Wind-turbine performance loaded directly from `csvs/wind_turbine_summary.csv`. RMSE is reported in kW; NLL is evaluated on standardized targets and is not defined for the deterministic model.

Model	K	Train NLL	Test NLL	Train RMSE	Test RMSE
Deterministic	—	N/A	N/A	383.646088	389.205292
Single Gaussian	1	-0.548644	-0.558781	386.383606	391.393005
MDN	2	-1.097666	-1.077309	392.833405	399.636749
MDN	3	-1.163613	-1.156921	390.928741	396.182831
MDN	4	-1.290105	-1.295361	391.118164	397.191193
MDN	5	-1.227277	-1.233597	393.009216	398.829529

5 Conclusion

Summarize when the MDN improves on deterministic and single-Gaussian regression, identify the best-supported mixture count, state limitations, and give one concrete next experiment.

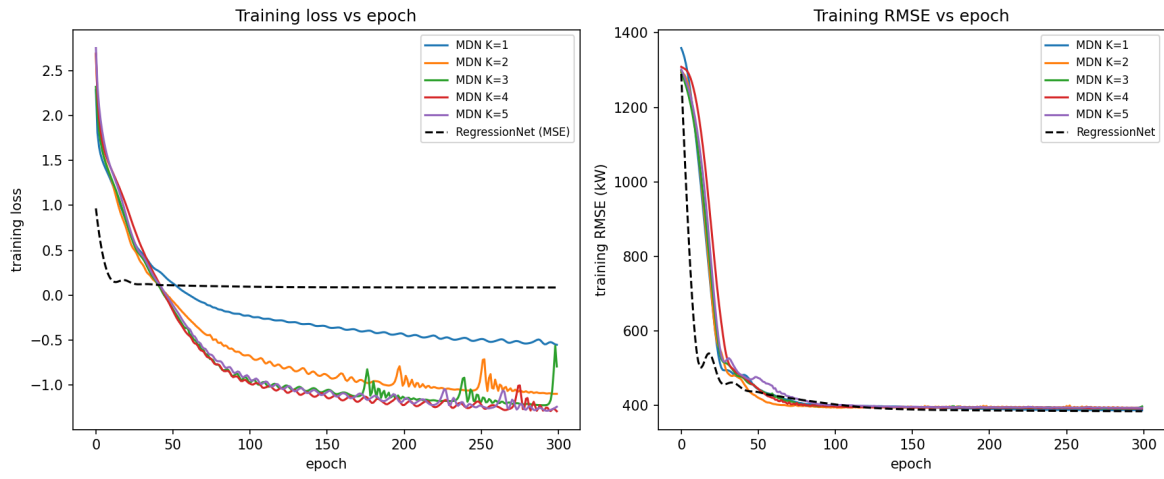


Figure 6: Training and validation/test behavior for the realistic-data models, including the selected checkpoint or stopping point.